

- Q.27 What are industrial automation tools?
- Q.28 What is the PLC Scan Cycle?
- Q.29 What is the Redundant System?
- Q.30 Draw the block diagram of PLC
- Q.31 Explain briefly the Remote Terminal Unit.
- Q.32 Explain briefly about various functional blocks of DCS.
- Q.33 Explain comparison instruction of PLC like equal , not equal, greater, greater than equal to.
- Q.34 What is Real Time Clock (RTC) in a PLC?
- Q.35 What are the main Functions of SCADA System?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What is Distributed Control System? Explain DCS System Layout?
- Q.37 What is SCADA? Explain the block diagram of SCADA system?
- Q.38 Write a short note on following
- AC drive for speed
 - Difference between DCS and SCADA

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6th Sem / Eltx Subject:- Industrial Automation

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Explain No instruction in Ladder Logic Programming
- Normally -open
 - Null-open
 - Normal-object
 - none of above
- Q.2 Expand PLC
- Program Less control
 - Program Logic Controller
 - Program Left control
 - None of the above
- Q.3 The main objective of a process control is
- to control electrical parameters
 - to control mechanical parameters
 - to control optical parameters
 - to control physical parameters
- Q.4 The PLC was invented by.
- Bill Gates
 - Dick Morley
 - Bill Landis
 - Tod Cunningham

- Q.5 Which one of the following is not a PLC manufacturer.
- a) Siemens b) Panasonic
c) Microsoft d) ABB
- Q.6 Abbreviate SFC?
- a) Sequential Function Charts
b) Serial Function Charts
c) Short Function Charts
d) None of these
- Q.7 Function Block Diagram is a type of _____?
- a) PLC Language
b) Block Diagram of a module
c) Block Diagram of a PLC model
d) None of these
- Q.8 _____ full form of DCS
- a) Digital computer system
b) Distributed control system
c) Both
d) None
- Q.9 The acronym PLC stands for
- a) Pressure Load control
b) Programmable Logic Controller
c) Pressure Loss Chamber
d) PID Loop Controller
- Q.10 What type of power supply is typically used for a PLC?
- a) AC b) DC
c) AC and DC d) Solar

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is Rung and Rail in PLC?
- Q.12 Expand HMI
- Q.13 Expand RTC
- Q.14 What is scan time of a PLC?
- Q.15 _____ is the heart of the SCADA system.
- Q.16 What is the function of a PLC's communication protocol?
- Q.17 A solenoid is an example of output device. (True/False)
- Q.18 What is the full form of SCADA?
- Q.19 What is Foundation Fieldbus (FF)?
- Q.20 What is Modbus?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write any five application of PLC labelled
- Q.22 What are the types of PLC Timer?
- Q.23 Write advantages of PLC over relays.
- Q.24 What is the difference between a PLC and a DCS?
- Q.25 What is the function of a PLC's watchdog timer?
- Q.26 Explain the different programming languages of PLC?

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